Q1. JEE Main 2026 (23 January Shift 1)

Four persons measure the length of a rod as $20.00 \mathrm{~cm}, 19.75 \mathrm{~cm}, 17.01 \mathrm{~cm}$ and $18.25 \mathrm{~cm}$. The relative error in the measurement of average length of the rod is :

(1) 0.08 (2) 0.06 (3) 0.18 (4) 0.24

Q2. JEE Main 2026 (21 January Shift 1)

In an experiment the values of two spring constants were measured as $k_1=(10 \pm 0.2) \mathrm{N} / \mathrm{m}$ and $k_2=(20 \pm 0.3) \mathrm{N} / \mathrm{m}$. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

(1) 1.33% (2) 1.67% (3) 2.67% (4) 2.33%

ANSWERS AND SOLUTIONS

1. (2) **2.** (2)

1. (2) Mean $= \frac{20.00+19.75+17.01+18.25}{4} = \frac{75.01}{4} = 18.7525$ cm.

Absolute errors: $|20.00 - 18.7525| = 1.2475$, $|19.75 - 18.7525| = 0.9975$, $|17.01 - 18.7525| = 1.7425$, $|18.25 - 18.7525| = 0.5025$.

Mean absolute error $= \frac{1.2475+0.9975+1.7425+0.5025}{4} = \frac{4.49}{4} = 1.1225$.

Relative error $= \frac{1.1225}{18.7525} \approx 0.06$.

2. (2) For parallel combination: $k_{eq} = k_1 + k_2 = 10 + 20 = 30$ N/m

For sum, absolute errors add:

$\Delta k_{eq} = \Delta k_1 + \Delta k_2 = 0.2 + 0.3 = 0.5$ N/m

Percentage error $= \frac{\Delta k_{eq}}{k_{eq}} \times 100 = \frac{0.5}{30} \times 100 = 1.67\%$